

HealthConnect Clinic Experience Lab

Week 6 Project Summary

Track: Data Analytics

My main goal for Week 6 was to build on the work I completed in Week 5 by looking more closely at some of the patterns connected with appointment no-shows.

I focused mainly on booking lead time, previous no-show history, appointment type and reminder effectiveness. I also went back to validate some of the important KPIs from Week 5 rather than starting the analysis again.

One of the clearest findings was booking lead time. The no-show rate increased from 29.47% for appointments booked within 0–7 days to 63.95% for appointments booked 31–60 days ahead. Previous no-show history also showed a strong pattern, with the no-show rate increasing from 46.30% for patients with no previous no-show to 70.33% for patients with three or more previous no-shows.

I also validated the overall no-show rate of 48.46%, confirmed that no-shows had a longer average booking lead time of 34.53 days compared with 24.52 days for attended appointments, and found that reminders were linked with a lower no-show rate.

I improved my Power BI dashboard by keeping the original HealthConnect Overview and adding a second page for the Week 6 advanced analysis. This helped me separate the deeper findings from the general dashboard without making the original page too crowded.

For my cross-track collaboration, I worked with someone from the Data Science track. I shared my more detailed booking lead-time analysis with her, and she independently validated the four groups and got exactly the same no-show rates. She then tested the refined feature in Logistic Regression and Random Forest. It did not lead to a major improvement in performance, but she kept the refined grouping because it was more detailed and better validated. She selected Logistic Regression as her final model because it maintained recall, which was important for identifying possible no-shows.

One of the main challenges this week was making sure the deeper analysis stayed consistent with the Week 5 KPIs and that the different no-show rates were calculated using the correct denominators.

My main contribution this week was turning some of the Week 5 findings into more detailed and useful insights that can support HealthConnect's decisions and also provide useful evidence for Data Science modelling.

The important decision I made was to keep the overall no-show rate of 48.46% as the main KPI while using the attended-versus-no-show rate only for the deeper comparisons, where cancelled appointments were excluded.

For Week 7, I plan to test the findings further, especially the relationship between booking lead time, reminder status and previous no-show history, and to see whether combining these factors gives clearer predictive or decision-making value.